

WEEK 3 PROJECT REPORT

Financial Data Import, Transformation & Time Intelligence

Dataset: Shopify Stock Data

Tool Used: Microsoft Power BI

1. Introduction

This project focuses on analyzing Shopify stock market data using Microsoft Power BI. The stock data was imported, cleaned and transformed using Power Query. A separate Date Dimension table was created to perform time-based analysis. DAX measures were also created to analyze stock prices, trading volume and daily returns.

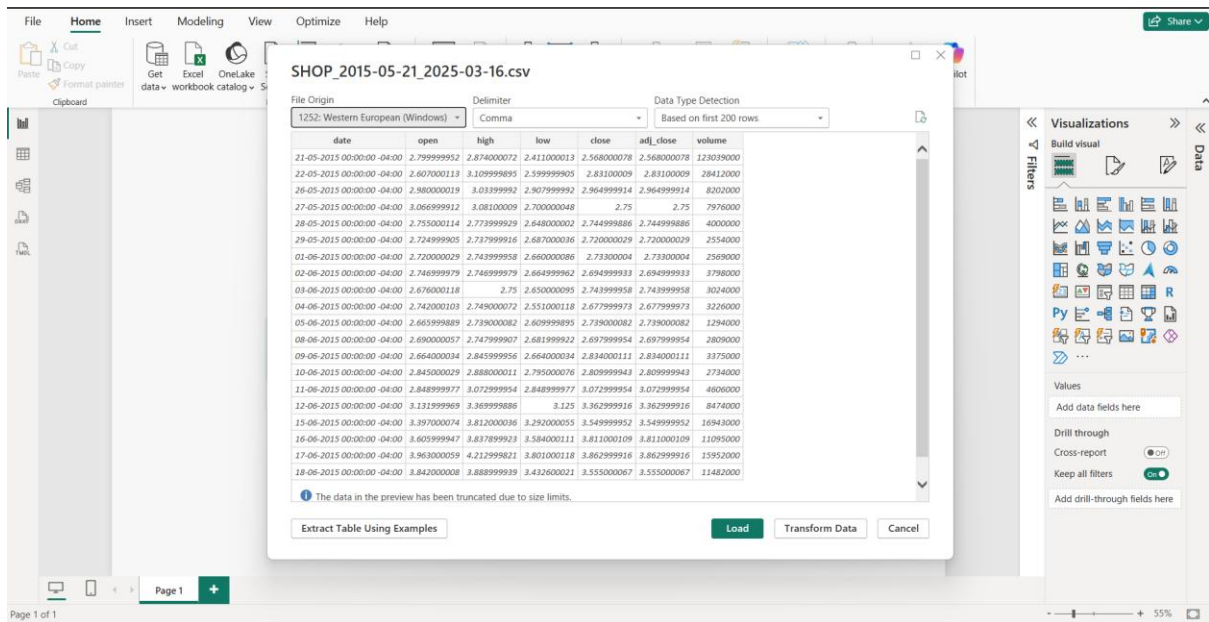
2. Objectives

The main objectives of this project are:

- To import and prepare Shopify stock data in Power BI.
- To clean and transform the stock market dataset using Power Query.
- To create a dedicated Date Dimension table.
- To establish a relationship between the Date table and stock data.
- To create DAX measures for stock price and trading activity.
- To implement time-intelligence calculations.
- To create a Year-Quarter-Month-Day hierarchy.
- To build a structured stock analysis dashboard.

3. Data Import

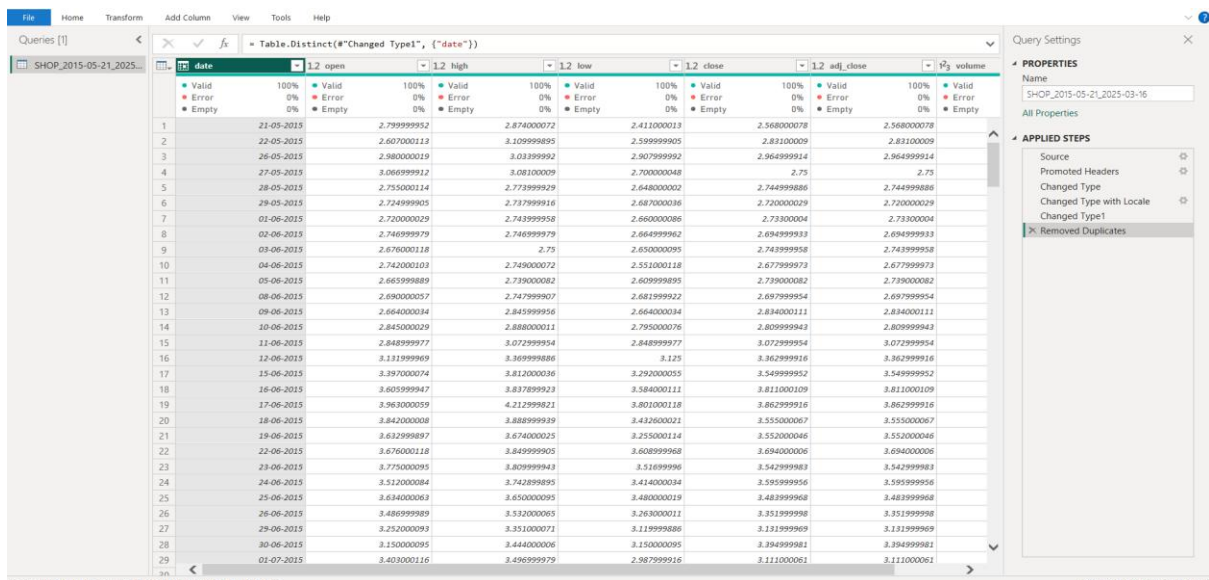
The Shopify Stock Data was imported into Power BI using the **Text/CSV** option. The dataset contains historical stock market information such as Date, Open Price, High Price, Low Price, Close Price and Trading Volume.



4. Data Cleaning and Transformation

The imported dataset was checked and cleaned using Power Query Editor. The data types of the columns were verified and the Date column was converted into the Date data type.

Missing values, duplicate records and incorrect values were also checked. The cleaned data was then loaded into Power BI for further analysis.

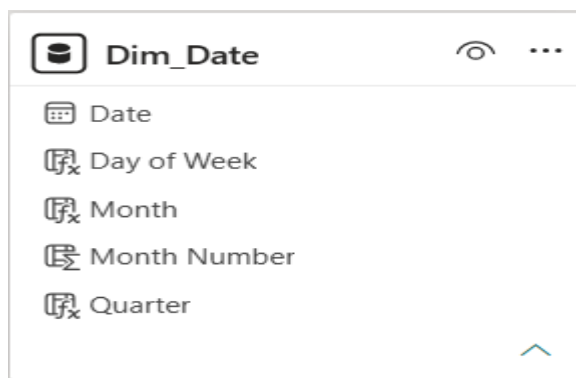


5. Date Dimension Table

A separate Dim_Date table was created using the CALENDARAUTO() DAX function. This table provides a structured date dimension for time-based analysis.

The following columns were created:

- Date
- Year
- Quarter
- Month
- Month Number
- Day
- Day of Week



6. Date Hierarchy

A custom date hierarchy was created using:

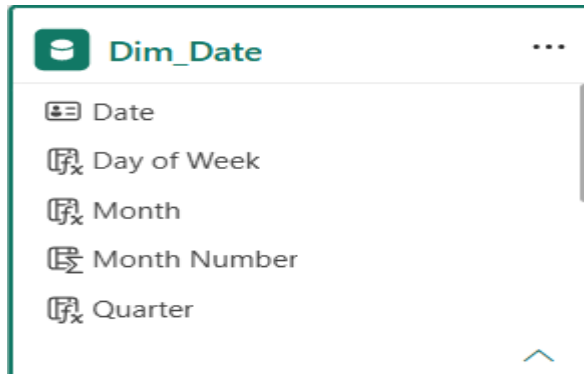
Year → Quarter → Month → Day

This hierarchy allows the stock data to be analyzed at different levels of time. Users can drill down from yearly information to quarterly, monthly and daily information.



7. Marking Dim_Date as Date Table

The Dim_Date table was marked as the official Date table in Power BI. The Date column was selected as the date column. This allows Power BI to correctly perform time-intelligence calculations.



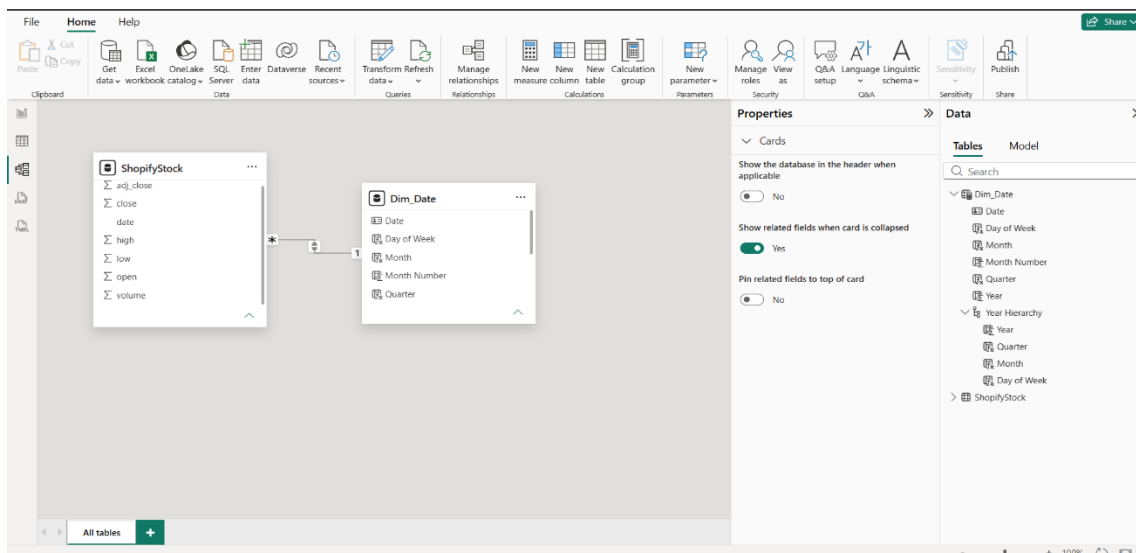
8. Data Model and Relationship

An active **one-to-many relationship (1:*)** was created between the Dim_Date table and the Shopify stock data table using the Date column.

The relationship is:

Dim_Date[Date] → ShopifyStock[Date]

The Dim_Date table acts as the Date dimension, while the Shopify stock table contains the stock price information.



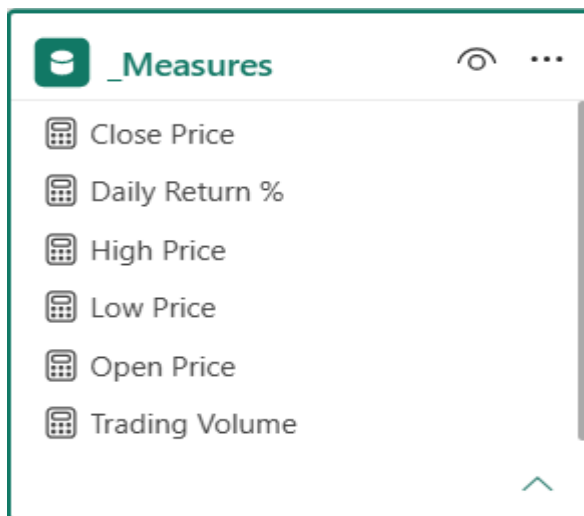
9. Financial DAX Measures

Several DAX measures were created to analyze the financial performance of Shopify stock.

The measures include:

- Open Price
- Close Price
- High Price
- Low Price
- Trading Volume
- Daily Return %

These measures were stored in a dedicated `_Measures` table.



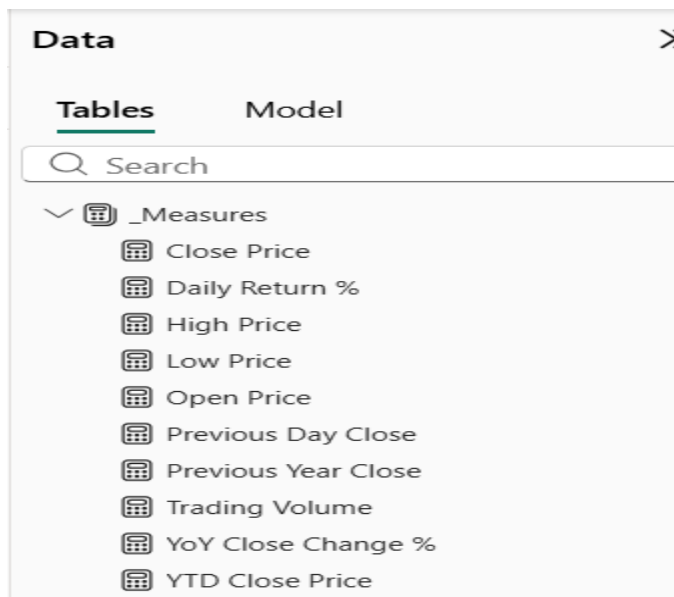
10. Time-Intelligence Measures

Time-intelligence calculations were created to compare stock performance across different time periods.

The following DAX functions were used:

- `DATEADD()` – to compare values with previous dates.
- `TOTALYTD()` – to calculate year-to-date values.
- `SAMEPERIODLASTYEAR()` – to compare the current period with the same period of the previous year.

These calculations help analyze changes in stock prices over time.



11. Stock Price Analysis

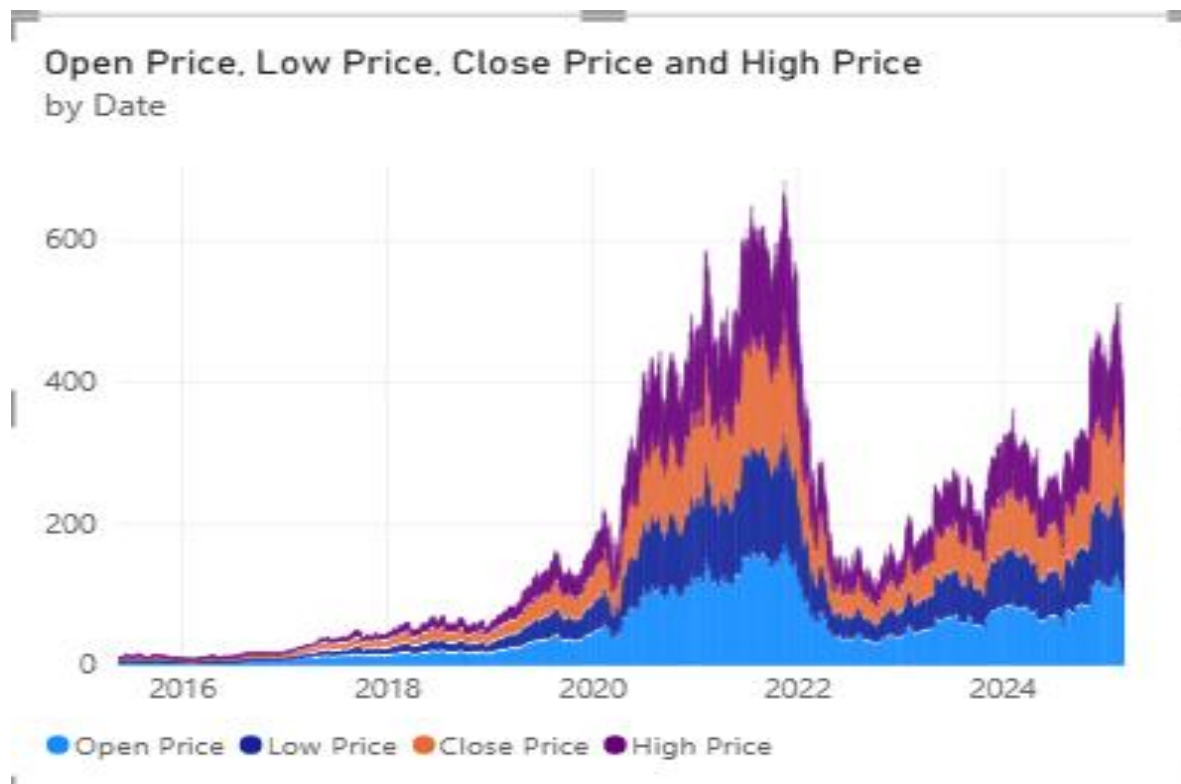
A line chart was created to display the Shopify closing price over time. The Date field was used on the X-axis and Close Price was used as the value.

This visualization helps identify changes and trends in the stock's closing price.



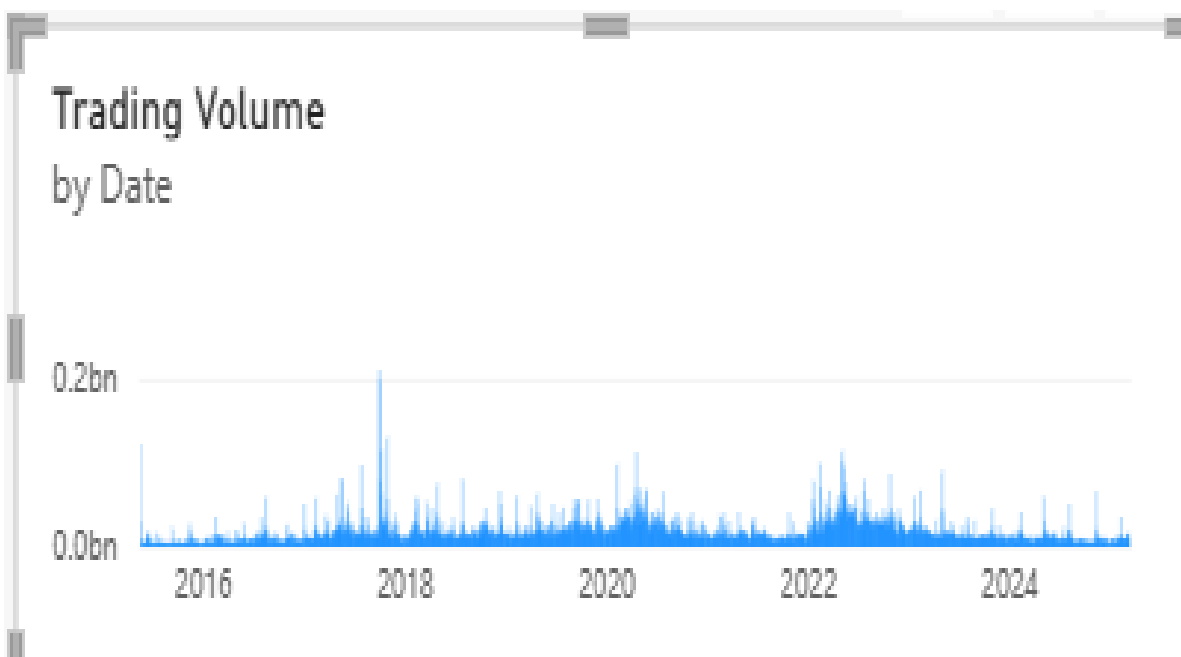
12. Open, High, Low and Close Price Analysis

A Stacked area chart was created to compare Open Price, High Price, Low Price and Close Price over time. This provides a detailed view of the daily movement of Shopify stock prices.



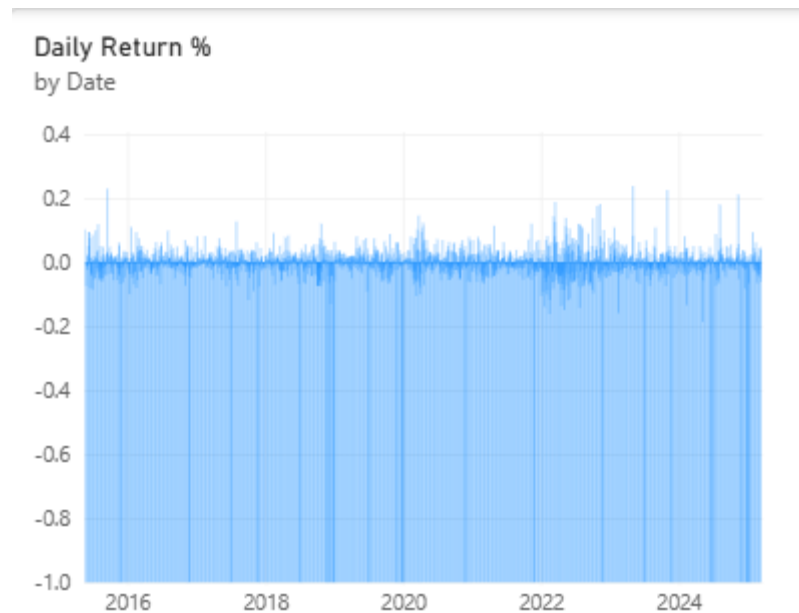
12. Open, High, Low and Close Price Analysis

A line chart was created to compare Open Price, High Price, Low Price and Close Price over time. This provides a detailed view of the daily movement of Shopify stock prices.



13. Trading Volume Analysis

A column chart was created to display Trading Volume over time. This visualization helps understand changes in trading activity during different periods.



14. Daily Return Analysis

The Daily Return Percentage measure was created to calculate the percentage change in the closing price from one trading day to the next.

A line chart was used to visualize the daily return percentage over time.

Date	Close Price	YoY Close Change %	YTD Close Price
21-05-2015 00:00:00	2.57	0.00	2.5
Total	48.46	0.03	110.6

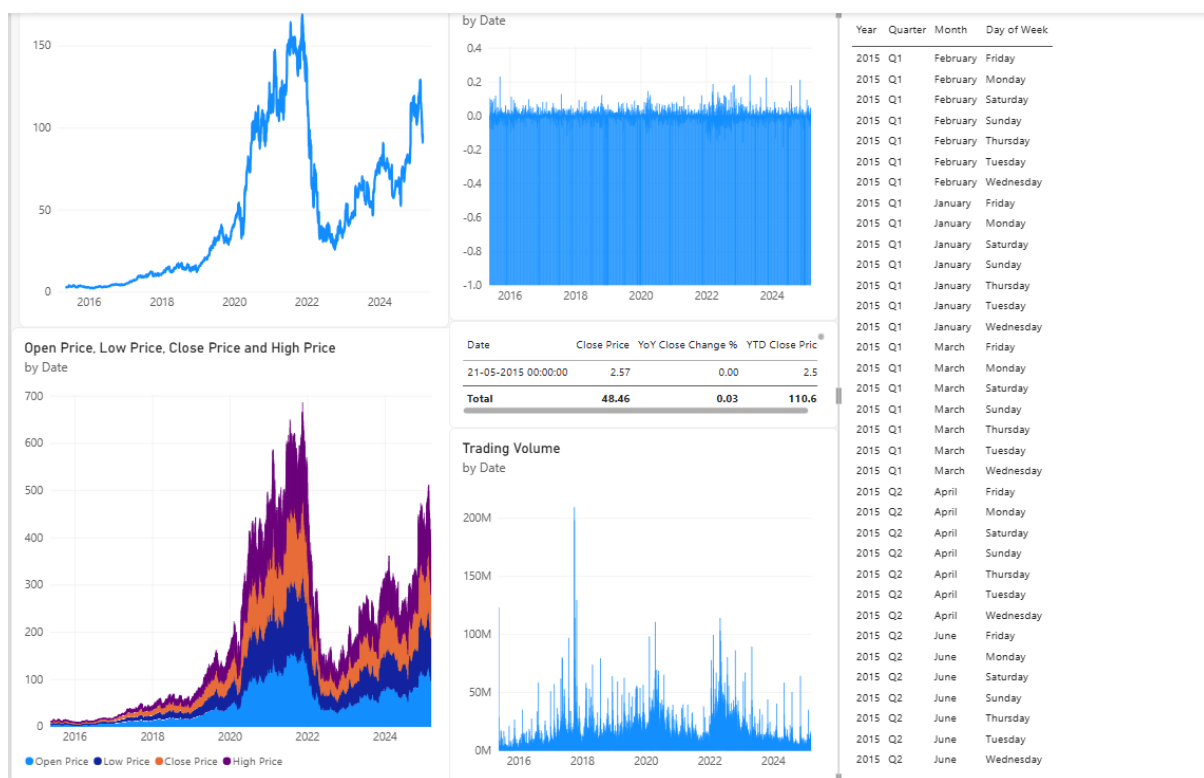
17. Final Dashboard

A final interactive Shopify Stock Dashboard was created by combining the financial measures and visualizations.

The dashboard includes:

- Open Price
- Close Price
- High Price
- Low Price

- Trading Volume
- Daily Return %
- Stock Price Trend
- Trading Volume Trend
- Time-based analysis
- Date filters
- Drill-down functionality



Explanation

The final dashboard provides an interactive view of Shopify stock performance. It combines key financial indicators, stock price trends, trading volume and time-based analysis in a single report page.

18. Conclusion

In this project, Shopify Stock Data was successfully imported and cleaned using Power Query. A dedicated Dim_Date table was created using DAX and marked as the official Date table. An active one-to-many relationship was established between the Date table and stock data.

Financial DAX measures were created for Open Price, Close Price, High Price, Low Price, Trading Volume and Daily Return Percentage. Time-intelligence functions such as DATEADD, TOTALYTD and SAMEPERIODLASTYEAR were also implemented.

Finally, different charts and an interactive dashboard were created to analyze Shopify stock prices and trading activity over time.